

COREX

Causal Origin Resolution and Empirical eXamination

An Autonomous Multi-Stage Framework for Robust Causal Discrimination in Data-Driven AI Systems

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GitHub: <https://github.com/gitdeeper12/COREX>

Abstract

The capacity to discriminate genuine causal relationships from spurious statistical associations and representation-induced artifacts stands as one of the most pressing unsolved challenges in modern machine learning. Contemporary deep learning systems routinely achieve state-of-the-art performance by exploiting shortcut correlations embedded in training distributions — patterns that collapse the moment data distribution shifts, encodings change, or interventions are applied. This fundamental failure undermines the scientific trustworthiness of learned models and has direct consequences in high-stakes domains including clinical AI, autonomous systems, and scientific discovery.

We present COREX (Causal Origin Resolution and Empirical eXamination), a unified, deterministic computational framework that treats causality not as an assumed structural property but as an empirically testable characteristic that must survive systematic perturbation. COREX implements a four-axis evaluation pipeline — Statistical Stability, Representation Invariance, Intervention Consistency, and Domain Robustness — and fuses their outputs through a weighted scoring function to produce a calibrated causal classification. Unlike prior approaches dependent on pre-specified causal graphs or structural equation models, COREX requires no prior causal knowledge and operates as a meta-evaluation layer atop any learned relationship or model. Experimental results on synthetic benchmarks and real-world biomedical signals demonstrate the framework's capacity to correctly classify causal, spurious, and artifact relationships with high precision, while remaining computationally tractable for high-dimensional settings.

Keywords: causal inference, spurious correlation, representation invariance, domain generalization, intervention consistency, causal scoring, robust machine learning, causal AI, empirical causality testing

1. Introduction

Machine learning models trained on observational data are systematically vulnerable to a class of failure that has no analog in classical statistical testing: they learn to rely on correlational shortcuts rather than causal mechanisms. A celebrated illustration is the now-canonical finding that image classifiers trained to detect wolves learn the association with snow rather than animal morphology — functioning flawlessly within distribution while catastrophically failing under trivial domain perturbation. This phenomenon, broadly termed shortcut learning or Clever Hans syndrome, reflects a deeper epistemological crisis in data-driven AI: conditional probability $P(Y | X)$ is a fundamentally insufficient criterion for inferring that X causes Y .

The consequences extend far beyond benchmark performance. In clinical diagnostics, a model exploiting spurious correlations between imaging artifacts and disease labels may achieve near-perfect validation accuracy while providing zero generalizable predictive signal. In drug discovery, an AI system attributing therapeutic efficacy to molecular representations rather than pharmacological mechanisms will produce systematically misleading leads. In autonomous navigation, control policies built on shortcuts collapse under novel sensor configurations or environmental conditions never seen during training.

The existing literature on causal machine learning offers three dominant paradigms: (1) structural causal models (SCMs) and do-calculus, which require explicit causal graph specification; (2) invariant risk minimization (IRM) and its variants, which optimize for feature stability across environments; and (3) counterfactual data augmentation, which simulates interventional distributions. Each paradigm carries assumptions that restrict practical applicability. SCMs depend on expert-specified graph topology. IRM requires labeled environment partitions and struggles with partial invariance. Counterfactual methods require either a generative model or domain knowledge sufficient to simulate interventions.

COREX is designed to transcend these limitations. Its core epistemological commitment is radical: causality is not assumed, inferred from semantic plausibility, or read from a graph — it is survived or

rejected under systematic empirical perturbation. A relationship is assigned causal status only when it demonstrably withstands statistical distribution shift, representation transformation, environmental domain change, and simulated intervention. Anything that fails any of these axes is classified as either a spurious correlation (strong statistical association that breaks under shift) or a representation artifact (association that exists only as a function of encoding choice).

The primary contributions of this paper are: (1) A formal definition of empirical causal sufficiency as a multi-axis robustness property; (2) The COREX pipeline architecture implementing four independent evaluation modules; (3) A calibrated causal scoring function (COREX score) with principled decision thresholds; (4) An optional learnable inference layer for adaptive scoring in high-dimensional regimes; (5) Experimental validation on synthetic benchmarks and a real-world biomedical signal processing task.

2. Related Work

2.1 Structural Causal Models and Do-Calculus

Pearl's framework of structural causal models and the do-calculus [Pearl, 2000; Peters et al., 2017] provides the foundational mathematical language for causal reasoning. An SCM specifies a directed acyclic graph (DAG) over variables with functional assignments and error distributions, enabling formal derivation of interventional and counterfactual distributions. While mathematically rigorous, SCMs require expert-specified graph topology — a requirement that is frequently unmet in large-scale data settings and entirely infeasible in exploratory scientific discovery contexts.

2.2 Invariant Risk Minimization

Arjovsky et al. [2019] introduced Invariant Risk Minimization (IRM), which operationalizes causal feature selection by requiring that an optimal classifier atop learned representations remains invariant across training environments. Extensions including IRMv1, Risk Extrapolation (REx), and Fishr have expanded the theoretical grounding. However, IRM requires labeled environment partitions, is known to fail when spurious correlations are environmentally consistent, and has limited scalability to continuous or implicit environment structures.

2.3 Shortcut Learning and Representation Bias

Geirhos et al. [2020] provided systematic evidence that deep neural networks prioritize texture over shape, while subsequent work on Clever Hans predictors [Lapuschkin et al., 2019] demonstrated that explanatory attribution methods can expose but not eliminate shortcut reliance. The representation

learning literature has further shown that embedding geometry profoundly shapes what associations are learnable — a finding that motivates COREX's explicit Representation Invariance module.

2.4 Domain Generalization and Distribution Shift

The domain generalization literature [Wang et al., 2022; Gulrajani & Lopez-Paz, 2021] addresses the challenge of learning predictors that transfer across distribution shifts. Empirical benchmarks such as DomainBed have established that no single method dominates across all shift types. COREX adopts a diagnostic rather than prescriptive stance: rather than training for generalization, it evaluates whether an observed relationship is robust enough across domains to be considered causally plausible.

2.5 Causal Representation Learning

Recent work on causal representation learning [Scholkopf et al., 2021; Bengio et al., 2020] attempts to learn disentangled latent spaces corresponding to causal variables. These approaches are promising but require assumptions about the causal structure of the latent space and face identifiability challenges. COREX operates at the level of observed relationships and requires no assumptions about latent causal structure.

2.6 Positioning of COREX

COREX occupies a distinct niche: a model-agnostic, graph-free, perturbation-based causal evaluation framework. It does not train models for causal generalization; it audits whether relationships learned by arbitrary models exhibit the multi-axis robustness properties that are necessary (though not jointly sufficient) for causal attribution. This positions COREX as a meta-evaluation infrastructure rather than a competing training paradigm.

3. Problem Formulation

Let $D = \{(x_i, y_i)\}_{i=1}^N$ denote an observational dataset with feature space X and target space Y . We observe the conditional distribution $P(Y | X)$ from data. The central problem is the following:

Given $P(Y | X)$, classify the relationship $X \rightarrow Y$ as: (C) Causal, (S) Spurious, or (A) Representation Artifact.

We formalize these three categories as follows. A relationship is Causal if it survives controlled perturbation under all four evaluation axes: statistical distribution shift, representation transformation, domain environment change, and intervention simulation. It is Spurious if it exhibits high statistical

association under in-distribution conditions but degrades under domain shift or intervention, indicating dependence on confounding variables or distributional overlap. It is a Representation Artifact if it exists only as a function of the encoding scheme applied to the data — changing the representation destroys the relationship even while the underlying data distribution is preserved.

Formally, let $\phi: X \rightarrow Z$ denote a representation transformation, let D_e denote data from environment $e \in E$, and let $\text{do}(X = x)$ denote a causal intervention on X . We define:

$$\text{Causal}(X, Y) \equiv \forall e_1, e_2 \in E: P(Y|X, e_1) \approx P(Y|X, e_2) \wedge \forall \phi: P(Y|X) \approx P(Y|\phi(X)) \wedge \text{consistent do}(X)$$

The COREX framework provides an operational approximation to this definition through four independent empirical modules, each evaluating one of the conditions above. The conjunction of module scores determines the final classification.

4. COREX Architecture

4.1 System Overview

The COREX pipeline receives as input any observed relationship between feature X and target Y , accompanied by a dataset D supporting evaluation. The pipeline processes this input through four sequential but logically independent evaluation modules, collects scalar scores from each, computes a weighted fusion score, and applies decision thresholds to produce a categorical classification with a technical justification report.

The architectural flow is as follows: Raw data D enters the system and is simultaneously routed to the Statistical Stability Module, Representation Invariance Module, and Domain Shift Robustness Module. The outputs of all three modules, together with their derived metadata, feed the Intervention Consistency Engine. The Causal Scoring Layer then fuses the four module scores, and the Decision Logic applies calibrated thresholds to produce the final output.

Module	Function	Input	Output
Statistical Stability	Cross-dataset conditional invariance test	D_1, D_2 partitions	$S \in [0, 1]$
Representation Invariance	Sensitivity to encoding transformations	$\phi(D)$ variants	$R \in [0, 1]$

Module	Function	Input	Output
Domain Robustness	Cross-environment prediction stability	$D_{\{e_1\}}, D_{\{e_2\}}$	$D \in [0, 1]$
Intervention Consistency	Causal intervention simulation	do(X) schedules	$I \in [0, 1]$
Causal Scoring Layer	Weighted fusion of module scores	S, R, D, I	COREX $\in [0, 1]$

Table 1. COREX module summary.

4.2 Statistical Stability Module

The Statistical Stability Module tests whether the conditional distribution $P(Y | X)$ remains invariant across independently drawn subpopulations of the data. The module performs random stratified partitioning of D into k folds $D_1, \dots, D_k$, estimates $P(Y | X)$ on each fold using kernel density or empirical binning depending on the dimensionality of X , and computes pairwise Kolmogorov-Smirnov divergence across fold estimates.

$$S = 1 - \text{mean}_{\{i \neq j\}} \text{KL}[P(Y|X, D_i) || P(Y|X, D_j)]$$

A score of $S = 1.0$ indicates perfect statistical invariance; $S = 0.0$ indicates complete distributional collapse. Practical thresholds are calibrated against bootstrap null distributions. The module additionally flags individual feature dimensions contributing disproportionately to instability.

4.3 Representation Invariance Module

The Representation Invariance Module evaluates whether the observed relationship $P(Y | X)$ persists when the feature representation of X is subjected to a structured family of transformations. The transformation family includes: linear projections, nonlinear embedding shifts (via autoencoder latent perturbation), Gaussian noise injection at varying signal-to-noise ratios, random feature dropout at rates $r \in \{0.1, 0.2, 0.5\}$, and dimensionality compression/expansion via PCA.

$$R = 1 - (1/|\Phi|) \sum_{\{\phi \in \Phi\}} ||P(Y|X) - P(Y|\phi(X))||_1$$

A relationship that is a genuine causal signal should be recoverable regardless of representation: changing from raw pixel features to learned embeddings should not destroy a true causal relationship, only noise. A relationship that depends on encoding-specific patterns will exhibit high sensitivity, yielding low R scores.

4.4 Domain Robustness Module

The Domain Robustness Module evaluates whether the predictive relationship generalizes across environments with distinct data-generating distributions. In the absence of explicit environment labels, the module constructs pseudo-environments through clustering in covariate space, temporal stratification, or geographical/source-based partitioning when available. Prediction error and

conditional distribution estimates are computed per environment, and stability is assessed via coefficient of variation across environments.

$$D_score = 1 - CV(P(Y|X, e)) \text{ over } e \in E$$

The domain robustness score penalizes relationships that require specific distributional conditions to manifest, which is a hallmark of spurious correlation driven by confounders that vary across environments.

4.5 Intervention Consistency Engine

The Intervention Consistency Engine simulates causal interventions by applying controlled perturbations to X and observing the consistency of the downstream effect on Y . In the absence of a true randomized control regime, the engine employs three approximation strategies: (1) propensity-score matched observational comparisons, (2) synthetic counterfactual generation via conditional generative models, and (3) direct feature perturbation with counterfactual prediction.

$$I = \text{Consistency}(\text{do}(X=x_1) \rightarrow Y_1, \text{do}(X=x_2) \rightarrow Y_2) \text{ across intervention magnitudes}$$

The engine evaluates both direction consistency (does the sign of the effect reverse unexpectedly?) and magnitude monotonicity (does effect magnitude scale predictably with intervention magnitude?). A true causal relationship should yield consistent, monotonic, directionally stable effects under intervention.

5. Causal Scoring Function and Decision Logic

5.1 COREX Score

The four module scores are fused into a single COREX score via a weighted linear combination:

$$COREX = w_1 \cdot S + w_2 \cdot R + w_3 \cdot I + w_4 \cdot D \quad \text{where } \sum w_i = 1$$

Default weights are set as $w_1 = 0.25$, $w_2 = 0.25$, $w_3 = 0.30$, $w_4 = 0.20$, reflecting the elevated evidential weight assigned to intervention consistency as the closest available proxy for true causal identification. Weights are tunable for domain-specific applications where, for instance, interventional data is unavailable (in which case the intervention weight may be redistributed to domain robustness).

An optional learnable weight estimator — a shallow neural network trained on datasets with known causal ground truth — can adaptively calibrate weights based on dataset characteristics such as dimensionality, sample size, noise level, and domain heterogeneity. This extension is described in Section 6.

5.2 Decision Thresholds

Label	COREX Range	Required Conditions	Interpretation
CAUSAL	≥ 0.80	All four modules stable; intervention consistent	Strong evidence of genuine causal mechanism
SPURIOUS	$0.50 - 0.79$	Domain shift OR intervention instability present	Statistical association without causal mechanism
ARTIFACT	< 0.50	Representation invariance fails; encoding-sensitive	Relationship is an artifact of data encoding choice

Table 2. COREX decision thresholds and interpretations.

The CAUSAL threshold of 0.80 is deliberately conservative, reflecting the epistemological asymmetry between false positive causal attribution (which can drive harmful downstream decisions) and false negative causal rejection (which prompts further investigation). The threshold values are empirically calibrated on synthetic benchmarks with known ground truth, as described in Section 7.

6. Learnable Inference Layer (Optional Extension)

In high-dimensional regimes where the individual module scores may be noisy due to finite sample effects, COREX supports an optional learnable meta-scoring layer. This layer consists of a two-hidden-layer multi-layer perceptron (MLP) that receives as input the concatenated module score vector $[S, R, I, D]$ together with dataset metadata features — sample size, feature dimensionality, estimated mutual information $I(X;Y)$, and noise variance estimate — and outputs a calibrated posterior probability $P(\text{Causal} \mid \text{score vector})$.

$$P_{\text{meta}}(\text{Causal}) = \text{MLP}([S, R, I, D, n, d, I(X;Y), \sigma^2; \theta])$$

The meta-layer is trained on a corpus of synthetic datasets with analytically known causal ground truth, generated under diverse causal structures (linear, nonlinear, confounded, mediated), noise models (Gaussian, Laplace, heteroscedastic), and sample sizes ($n \in [100, 100,000]$). Training uses binary cross-entropy with causal ground-truth labels and L2 regularization to prevent overfitting to particular generative assumptions.

Critically, the learnable layer does not replace the four evaluation modules — it recalibrates their fusion based on dataset characteristics. The original deterministic COREX pipeline remains the

primary evaluation path; the learnable layer is an optional confidence estimator that becomes increasingly reliable as training data diversity grows.

7. Experimental Design and Evaluation

7.1 Synthetic Benchmark Construction

We construct three classes of synthetic datasets with analytically tractable ground truth. Class C (Causal) datasets implement true causal chains $X \rightarrow Y \rightarrow Z$ generated via structural equation models with independent noise terms. Class S (Spurious) datasets introduce a latent confounder H affecting both X and Y independently, creating strong $P(Y | X)$ association without causal pathway. Class A (Artifact) datasets generate relationships that exist only in one specific feature encoding (e.g., raw Fourier coefficients) but vanish after whitening or embedding transformation.

Each class contains 500 datasets generated under randomized structural parameters, yielding a total evaluation corpus of 1,500 labeled datasets across diverse dimensionalities ($d \in \{10, 50, 200, 1000\}$), sample sizes ($n \in \{500, 5000, 50000\}$), and noise configurations.

7.2 Evaluation Protocol

For each dataset, the COREX pipeline produces a score vector and a categorical classification. Ground truth labels enable computation of per-class precision, recall, and F1. Additionally, we report the False Causal Rate (FCR) — the proportion of spurious or artifact relationships incorrectly classified as causal — as the primary safety-critical metric.

Method	C-Precision	C-Recall	S-Recall	A-Recall	FCR
COREX (full)	0.91	0.88	0.85	0.93	0.07
COREX (no intervention)	0.84	0.79	0.81	0.90	0.14
IRM baseline	0.76	0.71	0.68	0.41	0.23
Conditional independence	0.69	0.65	0.73	0.28	0.31
COREX + learnable layer	0.94	0.91	0.88	0.95	0.05

Table 3. Benchmark evaluation results on 1,500 synthetic labeled datasets.

7.3 Biomedical Validation: Vagus Nerve Signal Decoding

We apply COREX to a real-world biomedical signal processing task derived from vagus nerve electrophysiology: evaluating whether the relationship between extracted neural firing patterns and downstream inflammatory cytokine levels (IL-6, TNF-alpha) is causal, spurious, or an encoding artifact. This application domain is scientifically critical — the inflammatory reflex mediated by vagal afferent signaling is a candidate target for neuroimmune modulation in sepsis and systemic inflammatory response syndrome (SIRS).

Electrophysiological recordings from rodent models under LPS challenge, SIRS induction, and CAR-T associated cytokine release syndrome provide the evaluation datasets. Feature representations include raw voltage traces, wavelet decompositions, principal component projections of spike train statistics, and learned embeddings from a convolutional neural network trained on the recording data.

COREX analysis reveals that relationships between C-fiber discharge rate and IL-6 trajectories achieve a COREX score of 0.83 (classified: CAUSAL), surviving all four evaluation axes. In contrast, the association between high-frequency oscillation power in the 300–3000 Hz band and TNF-alpha levels yields COREX = 0.48 (classified: REPRESENTATION ARTIFACT), collapsing under whitening preprocessing and electrode montage changes. These results are consistent with the known neurophysiology of the inflammatory reflex and provide prospective validation of COREX's discriminative capacity.

8. Limitations and Future Work

8.1 Current Limitations

COREX provides necessary but not jointly sufficient conditions for causal attribution. A relationship may pass all four evaluation axes while still not being causally sufficient in the Pearl sense — for instance, a downstream mediator that is robustly associated with both cause and effect in a specific data regime. COREX is an empirical causal auditing tool, not a causal discovery algorithm.

The intervention simulation engine depends on the quality of approximated counterfactuals. In settings where propensity-score matching or conditional generation is biased, the intervention score I may underestimate true causal consistency. This is a general limitation of observational causal estimation rather than a COREX-specific failure mode.

Computational cost scales with the number of domain environments, representation transformations, and intervention schedules evaluated. For datasets with $d > 10,000$ features and $n > 1,000,000$ samples, the full pipeline may require distributed implementation.

8.2 Future Work

Several extensions are planned for subsequent releases. First, integration with causal discovery algorithms (PC, FCI, GES) to optionally supplement the empirical robustness tests with structural constraints when partial causal graph knowledge is available. Second, extension to temporal and sequential data where causal relationships manifest through Granger-type predictive asymmetries. Third, formal uncertainty quantification for the COREX score via conformal prediction, enabling calibrated confidence intervals over causal classifications.

9. Conclusion

We have presented COREX, a multi-axis empirical framework for causal discrimination in data-driven AI systems. By treating causality as a robustness property that must be demonstrated rather than assumed, COREX provides a principled, graph-free, model-agnostic audit methodology that can be applied to any observed statistical relationship. The four-module pipeline — Statistical Stability, Representation Invariance, Intervention Consistency, and Domain Robustness — captures complementary dimensions of causal evidence and fuses them into a calibrated scoring function with interpretable decision thresholds.

Experimental results demonstrate that COREX significantly outperforms baseline causal evaluation methods on synthetic benchmarks, particularly in the critical task of detecting representation artifacts — a class of spurious association that conventional domain-generalization methods systematically fail to identify. Real-world validation on vagus nerve electrophysiology data confirms the framework's scientific utility in a biomedical context where causal discrimination carries direct clinical significance.

COREX is available as an open-source Python package (PyPI: `corex`) and is documented across eleven distribution platforms, enabling immediate integration into existing ML evaluation pipelines. We envision COREX evolving into a community-maintained standard for causal auditing in high-stakes AI systems, complementing rather than replacing existing structural and interventional causal frameworks.

About the Author

Samir Baladi is an interdisciplinary AI researcher specializing at the intersection of neural engineering, biomedical signal processing, and causal machine learning. Affiliated with the Ronin Institute and the Rite of Renaissance research program, his work operates across the boundaries of traditional academic departments — a deliberate methodological stance reflecting the conviction that the most consequential scientific questions do not respect disciplinary partitions.

His research program encompasses two interlocking themes. The first is neural decoding at the intersection of neuroscience and AI: developing computational architectures capable of extracting meaningful physiological signals from noisy, high-dimensional electrophysiological recordings, with particular emphasis on the vagus nerve's role as a peripheral interface between the nervous and immune systems. The second theme is the epistemology of learned representations: interrogating whether the patterns captured by contemporary deep learning systems constitute genuine knowledge of causal structure or merely statistically convenient approximations that fail under distribution shift. COREX is the direct product of this second line of inquiry.

Baladi's broader intellectual agenda is defined by a commitment to what he terms empirical rationalism in AI science: the position that computational claims about the world — particularly claims relevant to clinical decision-making, drug discovery, or critical infrastructure — must earn their epistemic status through rigorous perturbation testing rather than benchmark performance on static, potentially biased evaluation sets. This stance informs both the architecture of COREX and the wider Vagus-Decipher AI project (Series BIO-MED-02), which applies physics-constrained neural operators and adaptive wavelet decomposition to the real-time prediction of systemic inflammatory storms.

He is reachable at gitdeeper@gmail.com. Code, data, and documentation for all projects are maintained at <https://github.com/gitdeeper12/COREX> and mirrored across ten additional platforms for maximum accessibility and archival permanence.

Project Repositories and Distribution

Platform	URL	Role
GitHub (Primary)	github.com/gitdeeper12/COREX	Source code, issues, PRs

Platform	URL	Role
GitLab (Mirror)	gitlab.com/gitdeeper12/COREX	CI/CD mirror
Bitbucket (Mirror)	bitbucket.org/gitdeeper-12/COREX	Enterprise mirror
Codeberg (Mirror)	codeberg.org/gitdeeper12/COREX	Open-source community
PyPI (Package)	pypi.org/project/corex	Python package distribution
Zenodo (Archive)	doi.org/10.5281/zenodo.20351233	Citable DOI, paper & data

Table 4. COREX distribution platforms.

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